

tf.convert_to_tensor

 [View source on GitHub \(https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/framework/tensor_conversion.py#L96-L163\)](https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/framework/tensor_conversion.py#L96-L163)

Converts the given `value` to a `Tensor`.

```
tf.convert_to_tensor(  
    value, dtype=None, dtype_hint=None, name=None  
) -> tf.Tensor (https://www.tensorflow.org/api\_docs/python/tf/Tensor)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Logistic regression for binary classification with Core APIs (https://www.tensorflow.org/guide/core/logistic_regression_core) • Extension types (https://www.tensorflow.org/guide/extension_type) • Matrix approximation with Core APIs (https://www.tensorflow.org/guide/core/matrix_core) • TensorFlow basics (https://www.tensorflow.org/guide/basics) • Distributed training with Core APIs and DTensor (https://www.tensorflow.org/guide/core/distribution)	• DeepDream (https://www.tensorflow.org/tutorials/generative/deepdream) • Custom training: walkthrough (https://www.tensorflow.org/tutorials/customization/custom_training_walkthrough) • Distributed training with DTensors (https://www.tensorflow.org/tutorials/distribute/dtensor_ml_tutorial) • Load a pandas DataFrame (https://www.tensorflow.org/tutorials/load_data/pandas_dataframe) • Scalable model compression (https://www.tensorflow.org/tutorials/optimization/compression)

This function converts Python objects of various types to `Tensor` objects. It accepts `Tensor` objects, numpy arrays, Python lists, and Python scalars.

For example:

```
>>> import numpy as np  
>>> def my_func(arg):  
...     arg = tf.convert_to_tensor(arg, dtype=tf.float32)  
...     return arg
```

```
>>> # The following calls are equivalent.  
...  
>>> value_1 = my_func(tf.constant([[1.0, 2.0], [3.0, 4.0]]))  
>>> print(value_1)  
tf.Tensor(  
  [[1. 2.]  
   [3. 4.]], shape=(2, 2), dtype=float32)  
>>> value_2 = my_func([1.0, 2.0], [3.0, 4.0])  
>>> print(value_2)  
tf.Tensor(  
  [[1. 2.]  
   [3. 4.]], shape=(2, 2), dtype=float32)  
>>> value_3 = my_func(np.array([1.0, 2.0], [3.0, 4.0]), dtype=np.float32))  
>>> print(value_3)  
tf.Tensor(  
  [[1. 2.]  
   [3. 4.]], shape=(2, 2), dtype=float32)
```

This function can be useful when composing a new operation in Python (such as `my_func` in the example above). All standard Python op constructors apply this function to each of their Tensor-valued inputs, which allows those ops to accept numpy arrays, Python lists, and scalars in addition to `Tensor` objects.

★ **Note:** This function diverges from default Numpy behavior for **float** and **string** types when **None** is present in a Python list or scalar. Rather than silently converting **None** values, an error will be thrown.

Args	
value	An object whose type has a registered Tensor conversion function.
dtype	Optional element type for the returned tensor. If missing, the type is inferred from the type of value .
dtype_hint	Optional element type for the returned tensor, used when dtype is None. In some cases, a caller may not have a dtype in mind when converting to a tensor, so dtype_hint can be used as a soft preference. If the conversion to dtype_hint is not possible, this argument has no effect.
name	Optional name to use if a new Tensor is created.

Returns
A Tensor based on value .

Raises	
TypeError	If no conversion function is registered for value to dtype .
RuntimeError	If a registered conversion function returns an invalid value.
ValueError	If the value is a tensor not of given dtype in graph mode.

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Last updated 2024-04-26 UTC.